

```
# Part A - Dictionaries

# 1. Create an empty dictionary
inventory = {}

# 2. Store the first product details in variables
ProductNo1 = "Product1"
ProductName1 = "Mobile Phone"
ProductQuantity1 = 5
ProductPrice1 = 20000
ProductReleaseYear1 = 2020

# 3. Add first product details to inventory
inventory[ProductNo1] = {
    "Product Name": ProductName1,
    "Product Quantity": ProductQuantity1,
    "Product Price": ProductPrice1,
    "Product Release Year": ProductReleaseYear1
}

# 4. Store the second product details in variables
ProductNo2 = "Product2"
ProductName2 = "Laptop"
ProductQuantity2 = 3
ProductPrice2 = 50000
ProductReleaseYear2 = 2022

# 5. Add second product details to inventory
inventory[ProductNo2] = {
    "Product Name": ProductName2,
    "Product Quantity": ProductQuantity2,
    "Product Price": ProductPrice2,
    "Product Release Year": ProductReleaseYear2
}

# 6. Display the products present in inventory
print("Products present in inventory:")
print(inventory)

# 7. Check if release years of both products are in inventory
if (inventory[ProductNo1]["Product Release Year"] == ProductReleaseYear1 and
    inventory[ProductNo2]["Product Release Year"] == ProductReleaseYear2):
    print("\nBoth release years are present in inventory.")
else:
    print("\nBoth release years are not present.")

# 8. Delete release year of both products
del inventory[ProductNo1]["Product Release Year"]
del inventory[ProductNo2]["Product Release Year"]
```

```
print("\nInventory after deleting release years:")  
print(inventory)
```

Products present in inventory:

```
{'Product1': {'Product Name': 'Mobile Phone', 'Product Quantity': 5, 'Product Price': 20000, 'Product Release Year': 2020}, 'Product2': {'Product Name':
```

Both release years are present in inventory.

Inventory after deleting release years:

```
{'Product1': {'Product Name': 'Mobile Phone', 'Product Quantity': 5, 'Product Price': 20000}, 'Product2': {'Product Name': 'Laptop', 'Product Quantity': 3
```

Part B - Tuples

Create a tuple of product prices

```
prices = (250, 300, 150, 400, 100, 350, 200)
```

Find and print the highest price

```
highest_price = max(prices)
```

```
print("Highest price:", highest_price)
```

Find and print the lowest price

```
lowest_price = min(prices)
```

```
print("Lowest price:", lowest_price)
```

Calculate and print the total sum

```
total = sum(prices)
```

```
print("Total sum of all prices:", total)
```

Convert tuple into sorted list in ascending order

```
sorted_prices = sorted(prices)
```

```
print("Sorted prices:", sorted_prices)
```

Try to modify an element in the tuple

```
# prices[0] = 500
```

Highest price: 400

Lowest price: 100

Total sum of all prices: 1750

Sorted prices: [100, 150, 200, 250, 300, 350, 400]

